

Zechuan Shi

Research Scientist | Experimental and Computational Biology | Functional Genomics, Perturbation Biology and AI for Science | tristanzeshi@gmail.com | GitHub: <https://github.com/rootze>

SUMMARY

Experimental and computational biologist with 6+ years of experience in assay development, preclinical research, functional genomics, and AI-enabled biological discovery. Developed perturbation-modeling and bioinformatics tools, designed studies across 600+ animals and 11 transgenics models, and integrated multi-omic datasets exceeding 2.3 million cells. Experienced in translating biological questions into testable experiments, computational workflows, and mechanistic hypotheses. Author of 15+ peer-reviewed publications, including Nature Genetics, and Science Advances.

TECHNICAL SKILLS

Experimental and Perturbation Biology: Cell culture; cell-based ELISA; assay optimization; antibody titration and dose-response testing; siRNA and CRISPR perturbations; flow cytometry; RNA extraction; qRT-PCR; preclinical treatment-response studies; target-engagement and off-target analysis; biomarker and resistance-mechanism discovery

Molecular and Cellular Data: RNA-seq; single-cell and single-nucleus RNA-seq; ATAC-seq; histone-modification profiling; spatial transcriptomics; Perturb-seq; flow cytometry; proteomics; multimodal data integration

AI and Computational Biology: Perturbation-response modeling; gene-regulatory networks; biological foundation models; representation learning; variational autoencoders; graph-based learning; interpretable machine learning; biological model evaluation

Programming and Research Infrastructure: Python; R; Bash; SQL; Git; Linux; HPC; Nextflow; reproducible workflows; sparse genomic data processing; software testing and documentation

RESEARCH AND EXPERIMENTAL EXPERIENCE

Postdoctoral Researcher / Graduate Student Researcher, University of California, Irvine, CA Sep 2020 – Present
Ph.D. completed Jun 2026; continued as Postdoctoral Researcher

Focus: Computational genomics and multi-omics modeling for therapeutic target discovery and perturbation analysis

- Developed and applied machine-learning and network-based methods for gene-regulatory inference, cell-state modeling, and therapeutic target prioritization.
- Integrated single-nucleus RNA-seq, chromatin-accessibility, histone-modification, and spatial-transcriptomic data across 2.3 million cells to identify disease-associated regulatory programs and candidate therapeutic interventions.
- Built reproducible Python and R research software and HPC workflows for multimodal biological datasets, including sparse matrices with millions of cells and hundreds of thousands of genomic features.
- Worked with experimental, clinical, and computational collaborators to translate model-derived predictions into biological hypotheses, validation strategies.

Biotech Investment Fellow, BlueRun Ventures – Beijing, China (Remote) Nov 2021 – Jan 2022

- Evaluated early-stage biotech companies in gene therapy, drug discovery, and computational biology, assessing technology feasibility, therapeutic potential, and development risks.
- Translated complex scientific innovations into actionable insights for investment decisions, analyzing competitive landscapes and communicating findings to cross-functional stakeholders.

Preclinical Research Specialist, University of Pennsylvania – Philadelphia, PA Jul 2018 – Jul 2020

Pharma Academic Partnership for Recurrent Breast Cancer Drug Development

- Designed and executed treatment-response studies involving 600+ mice across 11 HER2-positive transgenic breast cancer models.
- Performed mouse dosing, tumor measurements, tissue collection, necropsy, sample processing, and biobanking.
- Conducted cell culture, siRNA perturbation, RNA/DNA extraction, PCR/qPCR, Western blotting, immunostaining, and flow cytometry.
- Evaluated therapeutic response, target engagement, off-target effects, biomarkers, and resistance mechanisms using molecular and phenotypic readouts.
- Optimized experimental protocols and resolved assay, sample-quality, and data-quality issues.

- Integrated experimental and RNA-seq results to prioritize targets and guide follow-up studies with academic and pharmaceutical collaborators.

Research Intern, Gracomics LLC Laboratory – Kirksville, MO

Jan 2016 – May 2016

Cell-based ELISA Diagnostic Kit Development

- Developed and optimized a cell-based ELISA assay for preclinical research and diagnostic applications.
- Evaluated assay performance across multiple cell lines to assess robustness, reproducibility, and biological applicability.
- Refined experimental conditions and assay parameters to improve signal quality and reliability.
- Contributed to the development and validation of an ELISA-based diagnostic kit.

Research Fellow, University of Pennsylvania – Philadelphia, PA

May 2013 – Aug 2013

- Investigated stromal fibroblast-breast cancer interactions driving resistance to radiation-induced DNA damage.
- Identified STAT1 and NOTCH3 activation as a potential mechanism of stromal-mediated treatment resistance.
- Conducted antibody titration and dose-response experiments to define an effective interferon-neutralizing antibody range.
- Analyzed molecular and cellular readouts to guide follow-up experiments on resistance mechanisms.

SELECTED PROJECTS

COMPACT – In Silico Gene Module Perturbation Framework

Lead Developer | Ph.D. Thesis Project

- Developed a network-based perturbation modeling framework for predicting how targeted gene-module interventions reshape single-cell states and regulatory networks.
- Designed evaluation strategies using transcriptome-wide response, cell-state transition, network effects, robustness analysis, and comparison with observed perturbational data.
- Applied the framework across neurodegeneration and oncology datasets to prioritize candidate interventions and generate experimentally testable hypotheses.
- Assessed model robustness, biological generalizability, and interpretability using regulatory-network, motif, and multi-omic evidence.

Operon – AI-Powered Bioinformatics Desktop Platform

Contributor | Open-source academic software

- Developed a cross-platform application pairing Claude with published bioinformatics workflows for single-cell, RNA-seq, ATAC-seq, GWAS, and proteomics analysis.
- Built AI-assisted modes for autonomous analysis, workflow planning, PubMed-grounded question answering, and scientific report generation.
- Integrated 180+ curated protocols and enabled laboratories to convert existing scripts into reusable workflows deployable locally or on HPC systems.
- Designed the platform to help experimental biologists analyze their own data and enable bioinformaticians to test multiple hypotheses in parallel.

hdWGCNA – Network Analysis for High-Dimensional Transcriptomics

Core Contributor | Open-source R package

- Contributed to an R package for gene co-expression network analysis in single-cell transcriptomic data.
- Developed and applied workflows for network construction, module identification, eigengene analysis, hub-gene prioritization, and cross-dataset comparison.
- Supported documentation, testing, and reproducible analysis for a widely adopted open-source package with 480+ GitHub stars.

EDUCATION

Ph.D. in Biological Sciences, University of California, Irvine

Jun 2026

Research Areas: Computational modeling, Machine Learning, Genomics, GWAS, Single-cell RNA/ATAC/Spatial

Master of Science in Biotechnology, Johns Hopkins University, GPA: 3.92/4.0

May 2018

SELECTED PUBLICATIONS (15+)

- **Shi, Z.***, Morabito, S.*, et al. (2026). In silico module perturbation analysis unlocks a functional understanding of the dynamic gene networks in single-cell data. (Manuscript in preparation)
- **Shi, Z.**, Das, S., Morabito, et al. (2025). Single-nucleus multi-omics identifies shared and distinct pathways in Pick's and Alzheimer's disease. **Science Advances**.
- Fotio, Y., Al Masri, S., **Shi, Z.**, et al. (2026). Metabolic reallocation in spinal cord oligodendrocytes drives chronic pain via neuronal β -amyloid production. (Manuscript in press; **Science Translational Medicine**)
- Miyoshi, E., Morabito, S., Henningfield, C.M. [et al., including **Shi, Z.**]. (2024). Spatial and single-nucleus transcriptomic analysis of genetic and sporadic forms of Alzheimer's Disease. **Nature Genetics**.
- Tiwari, V., Prajapati, B., Asare, Y. [et al., including **Shi, Z.**]. (2024). Innate immune training restores pro-reparative myeloid functions for remyelination in the aged central nervous system. **Immunity**, 57(9), 2173–2190.e8.
- O'Shea, T.M., Ao, Y., Wang, S. [et al., including **Shi, Z.**]. (2024). Derivation and transcriptional reprogramming of border-forming wound repair astrocytes after spinal cord injury or stroke in mice. **Nature Neuroscience**.
- **Shi, Z.***, Das, S.*, Morabito, S., Miyoshi, E., & Swarup, V. (2022). Protocol for single-nucleus ATAC sequencing and bioinformatic analysis in frozen human brain tissue. **STAR Protocols**, 3(3), 101491.

SELECTED PRESENTATIONS

- "The new era of bioinformatics: How foundation models will rewrite the game in functional genomics". **NeuroBlitz**, Irvine, CA. Nov 2025. (Talk)
- "Recent advances in Perturb-seq". Single-cell Journal Club at **UCLA, Daniel Geschwind Lab**, Los Angeles, CA. Oct 2025. (Invited Talk)
- "In silico network reprogramming of oligodendrocytes reveals therapeutic potential in Alzheimer's disease". **Annual Meeting of American Society of Human Genetics**, Boston MA. Oct 2025. (**Reviewers' Choice Abstract Award**)
- "In Silico Module Perturbation Analysis unlocks a functional understanding of the dynamic gene networks in single-cell data". **Annual Meeting of American Society of Human Genetics**, Denver, CO. Nov 2024. (Talk)
- "Single-nucleus open chromatin accessibility landscape of Pick's and Alzheimer's disease". **Alzheimer's Association International Conference**, San Diego, CA. Jul-Aug 2022. (Talk)

FELLOWSHIP & AWARDS

James L. McGaugh Excellence Award (2026): Excellence in Research Award

ASHG Reviewers' Choice Abstract Award (2025): Awarded to the top 10% of submissions as scored by reviewers

Biological Sciences Travel Award (2023, 2024, 2025): University of California Conference Travel Award (\$900)

Bio2X Outstanding Trainee Award (2021-2022): BlueRun Ventures Bio2X Life Sciences Ideation Award

Cum Laude (2015): Graduating with Honor, TSU

Gerhardt Research Fellow (2015): a competitive research award (\$3,000)

Grants-in-Aid of Scholarship and Research (GIASR) (2014): a competitive research grant for the initial stage of an individual student research project (\$750)

President's Honorary Scholarship (2011-2015): an annual academic achievement scholarship, TSU (\$18,000)

TEACHING & LEADERSHIP EXPERIENCE

Program Manager, genPALs and UCI GPS – STEM Data Science Cohort – Irvine, CA Jul 2022 – Jun 2024

- Organized bi-weekly genomics and data science seminars, facilitating engagement between academic researchers and industry professionals.
- Developed and taught workshops on introductory machine learning and statistical methods for single-cell transcriptomic analysis.

Graduate Teaching Assistant, University of California – Irvine, CA Mar 2022 – Mar 2024

Classes: Bio 38: Mind, Memory, and Brain; Bio 48: The Mind-Body Connection in the Neuroscience of Well-Being; N 138: Sex Influences on Brain; NEURBIO 227. Bioinformatics and Systems Biology

- Graded 200+ students' writing assignments, programming tasks and exams
- Facilitated in-class discussion sections and answered students' questions